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RESEARCH ARTICLE

Management of tomato late blight disease in high rainfall area of Northeast India

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Abstract

Tomato cultivation in Northeast India faces a formidable challenge from late blight, caused by *Phytophthora infestans* (Mont.) de Bary. Field experiments conducted during 2021–2023 have shed light on effective management strategies. Among these, copper oxychloride at 0.25% emerged as the most effective treatment, closely followed by copper hydroxide at 0.25%. An optimized spray schedule revealed that six canopy sprays of copper oxychloride at 6-day intervals, in conjunction with black polythene soil mulching, achieved the highest disease control at 92.04%. Five sprays offered 89.77% control, while four sprays provided 81.82% control. The pooled mean yield for copper oxychloride-treated tomatoes was a remarkable 9.25 t/ha, surpassing that of copper hydroxide treatments. Furthermore, an evaluation of eight germplasms under natural epiphytotic conditions identified Arka Abhed F_1 as resistant to late blight. Consequently, for effective management of late blight in high rainfall areas, it is recommended to cultivate Arka Abhed F_1 or implement six canopy sprays of copper oxychloride with soil mulching for susceptible varieties/hybrids. These findings offer a sustainable and potent strategy for combating late blight, ensuring healthier yields for tomato growers in the region.

Keywords Tomato · Late blight · Organic fungicides · Arka abhed F_1 · Northeast India

Introduction

Organic agriculture is gaining global importance, with significant growth observed in the market for organic products (Suja 2013). In a landmark achievement in 2016, Northeast India converted its entire cultivable land to organic certification (Gopi et al. 2020).

Tomato, revered for its nutritional and medicinal value, is often dubbed a 'Nutraceutical vegetable' (Afroz et al. 2009; Noureen et al. 2010). Thriving in warm seasons, it adapts well to both plains and hills. In Northeastern India, tomato cultivation during the rainy season has gained prominence due to high market prices compared to other periods (Babu 2006).

Despite its significance, tomato yields in India are hindered by the pervasive impact of diseases, notably late blight, which causes widespread devastation (Son et al. 2008). Globally, annual losses from late blight reach an estimated $170 billion, posing a critical threat to food security (Wu et al. 2012). In India, yield losses of up to 79% due to late blight have been recorded (Chowdappa et al. 2015). Addressing this disease is imperative for improving tomato productivity and ensuring food security.

Host resistance is a vital strategy against late blight, yet currently, no tolerant or resistant tomato cultivars are available worldwide (Islam et al. 2013). Consequently, farmers often rely on systemic fungicides for control, but these are not permissible in organic agriculture. This underscores the urgency to explore alternative approaches that align with organic principles and support sustainable tomato production.

In the realm of organic agriculture, where systemic fungicides are not viable, farmers turn to alternative methods such as plant-derived products, bio-control agents, and copper and sulfur-based fungicides to combat diseases like late blight in tomatoes. In the lush landscapes of Northeast India, characterized by abundant rainfall and high humidity,

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conditions are ideal for pathogen proliferation (Gopi et al. 2020). These environmental factors often render conventional disease management measures ineffective, posing significant challenges for agricultural practices in the region. The open cultivation of tomatoes exacerbates these challenges, making effective disease control a pressing concern. Against this backdrop, the current study endeavours to evaluate various approaches tailored to these unique environmental and agricultural conditions. By innovating with methods that harmonize with nature, the goal is to safeguard tomato crops and promote sustainable farming practices in this biodiverse region.

Materials and methods

Effect of different treatments under field conditions

Field experiments were conducted during 2021 and 2022 at the ICAR Research Farm, Arunachal Pradesh Centre, Basar (650 amsl). The planting was done in the second week of March in both years, using the tomato variety Pusa Ruby, which is highly susceptible to *P. infestans* (Singh et al. 2019). The field trials were laid out in a Randomized Block Design (RBD) with four replications in plots measuring 1.8 m × 1.8 m, with a spacing of 90 cm × 60 cm between rows and plants. During transplanting, 250 g of vermicompost per plant was applied in the planting hole and covered with a thin layer of soil, while additional doses of 150 g per plant were applied as top dressing at 60 DAT (Babu 2006). A total of five treatments were used, viz. copper oxychloride 50% WP @0.25%, copper hydroxide 77% WP @0.25%, metalaxyl 35% WS @0.1%, black polythene mulching and control with no spray (Table 1). Six sprays of the above treatments were administered immediately after the appearance of the disease at 6-day intervals. At physiological maturity, tomato fruits from each plot were harvested and weighed separately to determine fruit yield.

Standardization of spray schedule of copper oxychloride 50% WP

Field experiments were conducted during 2022 and 2023, with planting done in the second week of October in both years. The tomato variety Pusa Ruby was used. The field trials were arranged in a RBD with three replications in plots measuring 1.8 m ×1.8 m, with a spacing of 90 cm × 60 cm between rows and plants. Eight treatments were tested, involving different spray schedules of copper oxychloride 50% WP combined with black polythene (30 micron) mulching (Table 2). The canopy spray schedule was initiated immediately after the appearance of the disease, with sprays administered at 6-day intervals.

Field evaluation of tomato germplasm

Field experiments were conducted during 2022 and 2023, with planting done in the second week of October in both years. Eight commercial hybrids/cultivars of tomato were used in the experiment (Table 3). The field experiment was laid out in a Randomized Block Design (RBD) with three replications. Each plot measured 1.8 m x 1.8 m, with rows spaced 90 cm apart. The distance from plot to plot was 50 cm, and from plant to plant was 60 cm. Three plants per row and six plants per plot were maintained. Common cultural practices were followed throughout the cropping period.

Disease severity

The severity of late blight disease was assessed visually on the leaves, stems, and fruits of all plants in each replication, using the rating scale by Irzhansky and Cohen (2006). Assessments were made when all plants in the control treatment were infected with late blight under natural epiphytotic conditions. The severity grades were then converted into a percentage disease index (PDI) for analysis, using the formula provided by Wheeler (1969).

Table 1 Effect of different treatments on the severity of late blight and yield of tomato

Treatment	Disease severity (PDI)				Yield (t/ha)		
	2021	2022	Pooled	% decrease over control **	2021	2022	Pooled
Copper oxychloride 50% WP (0.25%)	34.96	35.66	35.31	63.89	09.75	08.75	9.25
Copper hydroxide 77% WP (0.25%)	38.75	39.75	39.25	59.86	08.50	08.00	8.25
Metalaxyl 35% WS (0.1%)	32.41	30.83	31.62	67.66	10.25	10.75	10.50
Black polythene (30 micron) soil mulching	60.27	61.95	61.11	37.50	05.37	04.63	5.0
Control	96.86	98.70	97.78	-	00.45	00.40	0.42
LSD ($P=0.05$)	3.36	2.25	01.86	-	00.69	01.41	0.73

**The per cent decrease over control: Disease reduction (%) = $\frac{\text{PDI in control} - \text{PDI in treatment}}{\text{PDI in control}} \times 100$

Table 2 Field efficacy of copper oxychloride 50% WP and black polyethene mulching against late blight of tomato

Treatment	Disease severity (PDI)			% decrease over control **
	2022	2023	Pooled	
Single canopy spray of copper oxychloride 50% WP (0.25%) at initial symptoms & black polythene (30 micron) soil mulching	49.50	52.72	51.11	47.73
Two canopy sprays of copper oxychloride 50% WP (0.25%) at 6-day intervals & black polythene (30 micron) soil mulching	31.24	33.20	32.22	67.05
Three canopy sprays of copper oxychloride 50% WP (0.25%) at 6-day intervals & black polythene (30 micron) soil mulching	20.92	21.30	21.11	78.41
Four canopy sprays of copper oxychloride 50% WP (0.25%) at 6-day intervals & black polythene (30 micron) soil mulching	17.36	18.20	17.78	81.82
Five canopy sprays of copper oxychloride 50% WP (0.25%) at 6-day intervals & black polythene (30 micron) soil mulching	08.80	11.20	10.00	89.77
Six canopy sprays of copper oxychloride 50% WP (0.25%) at 6-day intervals & black polythene (30 micron) soil mulching	07.58	07.98	07.78	92.04
Black polythene (30 micron) soil mulching	60.27	61.95	61.11	37.50
Control	96.86	98.70	97.78	-
LSD ($P=0.05$)	**09.22**	**05.90**	**09.28**	**-**

**The per cent decrease over control: Disease reduction (%) = $\frac{\text{PDI in control} - \text{PDI in treatment}}{\text{PDI in control}} \times 100$

Table 3 Late blight disease severity and disease reaction of tomato germplasm under natural epiphytotic conditions

Germplasm	Disease severity (PDI)			Disease reaction
	2022	2023	Pooled	
Arka Samarat F_1	73.33	66.67	70.00	HS
Arka Rakshak F_1	86.67	73.33	80.00	HS
Arka Abhed F_1	13.33	18.33	15.83	R
Arka Apeksha F_1	66.67	71.67	69.17	HS
Arka Aditya F_1	93.33	80.00	86.67	HS
Arka Vishesh F_1	80.00	75.00	77.50	HS
Kashi Adarsh	100.00	93.33	96.67	HS
Pusa Ruby (susceptible check)	100.00	100.00	100.00	HS
LSD ($P=0.05$)	15.52	19.05	13.47	-

*Whereas, R = resistant and HS = Highly susceptible

Statistical analysis

The data were analyzed using WASP 2.0 software, developed by the Central Coastal Agricultural Research Institute, Goa (India). Data were subjected to analysis of variance (ANOVA) using the Least Significant Difference (LSD) test. Percentage data were converted to arc sine value before analysis, following the methodology outlined by Gomez and Gomez (1984).

Results

Effect of different treatments under field conditions

Over a two-year study, copper-based fungicides demonstrated remarkable efficacy in managing late blight severity (Table 1). In 2021, copper oxychloride @ 0.25% emerged as the most potent treatment, reducing disease severity to 34.96%, while copper hydroxide @ 0.25% followed closely with a severity of 38.75%. The trend continued in 2022, with copper oxychloride maintaining its superior performance at 35.66% disease severity, and copper hydroxide again ranking second with 39.75%. Averaging the data from both years, copper oxychloride consistently provided the lowest late blight severity at 35.31%, closely trailed by copper hydroxide at 39.25%. These results underscore the effectiveness of copper oxychloride and copper hydroxide in significantly mitigating late blight, making them reliable choices for disease management.

The yield data further highlights the benefits of using copper-based fungicides in tomato cultivation. Plots treated with copper oxychloride achieved the highest average yield of 9.25 t/ha, followed by those treated with copper hydroxide, which produced a respectable average yield of 8.25 t/ha (Table 1).

Standardization of spray schedule of copper oxychloride 50% WP

An optimized spray schedule has revealed the remarkable efficacy of copper oxychloride in conjunction with black polythene soil mulching for controlling late blight in tomatoes. Six canopy sprays at 6-day intervals achieved an impressive 92.04% disease control, with five sprays offering 89.77% control, and four sprays providing 81.82% control (Table 2).



Fig. 1 View of highly susceptible (R_3T_8; Pusa Ruby) and resistant (R_1T_3; Arka Abhed F_1) germplasm of tomato

Field evaluation of tomato germplasm

Eight germplasms were evaluated against late blight disease under natural epiphytotic conditions. Among them, Arka Abhed F_1 exhibited resistance to the late blight disease (Table 3; Fig. 1).

Discussion

The effectiveness of copper fungicides in managing various plant diseases is well-documented (Teviotdale et al. 1989; Patel et al. 2014), with their pathogen suppression attributed to the non-specific denaturation of proteins (Ware and Whitacre 2004). The potent copper ions halt the growth of fungal spores, with Funguran (cupric hydroxide) demonstrating significant efficacy against late blight in tomatoes (Alexandrov 2011). Similarly, Platt et al. (1998) found that treating potato plots with copper oxychloride reduced foliar and tuber blight, leading to increased tuber yields. Our research supports these findings, showing a 64.51% reduction in late blight severity in tomatoes following six sprays of copper oxychloride at 0.25% (Gopi et al. 2020). Ghazanfar et al. (2010) also reported the effectiveness of copper hydroxide in reducing late blight incidence in potatoes. Furthermore, good foliage coverage with copper fungicides has been shown to control late blight in organic potato crops (Manuela and Hermeziu 2014). Narayan Bhat et al. (2007) demonstrated that copper sulphate significantly reduced the lesion size of *P. infestans* in potatoes. These findings underscore the potential of copper-based treatments as a robust strategy for managing late blight and enhancing crop yields. Notably, the lowest disease severity was observed in plots treated with the standard fungicide metalaxyl @ 0.1%. Metalaxyl, a systemic fungicide, has been extensively used to control late blight in potatoes and tomatoes, as well as other diseases caused by oomycete fungi (Shashidhara et al. 2009; Gopi et al. 2020). These findings suggest that while copper oxychloride and copper hydroxide significantly enhance yield, metalaxyl remains the most effective option for minimizing disease severity.

Evaluating tomato germplasms for resistance reveals the persistent challenge in breeding highly resistant varieties/hybrids. Studies by Islam et al. (2001) and Khalid et al. (2012) underscore this difficulty, identifying resistant, moderately resistant, and tolerant lines but no highly resistant genotypes.

The findings of the present study reveal that for effective organic management of late blight, cultivating Arka Abhed F_1 or employing six sprays of copper oxychloride at 0.25%, coupled with black polythene (30 micron) soil mulching, for susceptible varieties/hybrids in high rainfall areas of northeast India. Crucially, no phytotoxic symptoms were observed with copper fungicides, reinforcing their role as a sustainable and effective strategy. Given their protectant nature, copper fungicides must be applied before infection or immediately after the appearance of the disease. To maintain coverage and prevent disease establishment, these fungicides should be reapplied as plants grow. In wet weather, copper fungicides should be applied as soon as the disease is observed or when conditions are favourable for disease development. These findings offer a sustainable and potent strategy for combating late blight, ensuring healthier yields for tomato growers in the region.

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collection of germplasm, H. Kalita: Helped in research works and collection of germplasm.

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Declarations

Conflict of interest On behalf of all authors, the corresponding author states that there is no conflict of interest.

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